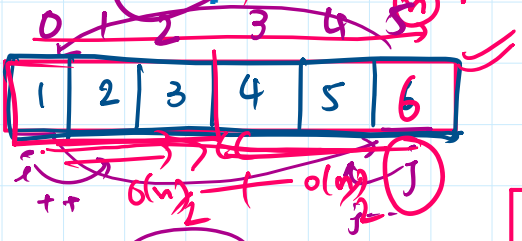
Two pointers
29 May 2026 20:54

Reverse an array

1



6 → (n-1) 2(6-1) = 5

plan

two pointers

variables → i, j

space complexity → O(n)

time complexity → O(n)

6 2 3 4 5 1

for

6 5 3 4 2 1 → while

(i <= j) → [not swapped]

6 5 4 3 2 1

swap(— —);

i++;

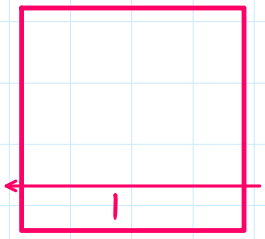
j--;

→ }

2 (stacks) → DS

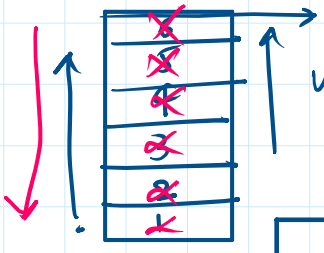
LIFO → Last in first out

functions : pop(), peek(), top(), push()

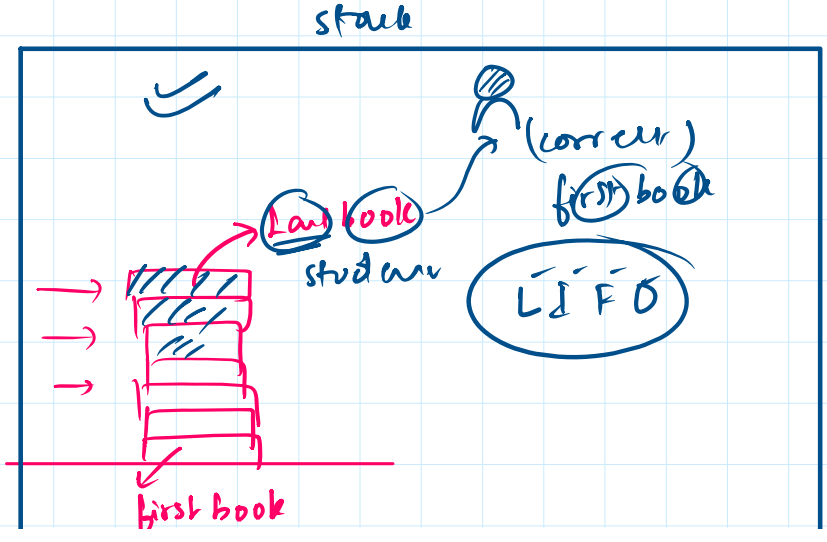


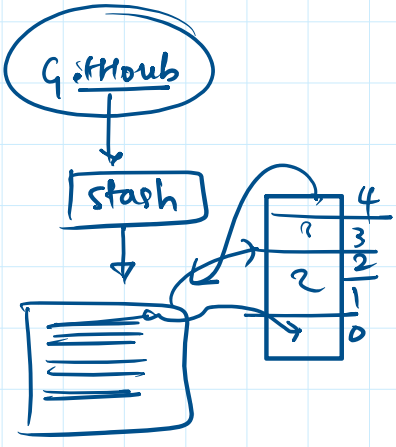
1 2 3 4 5 6 → Horizontally

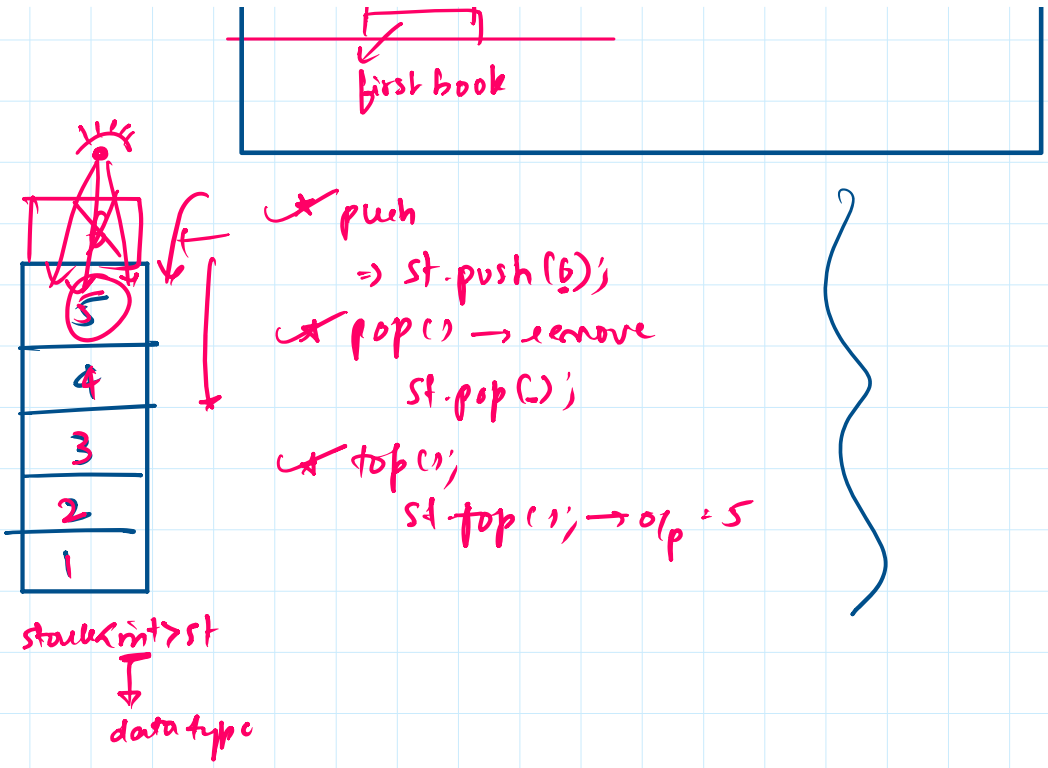
for :



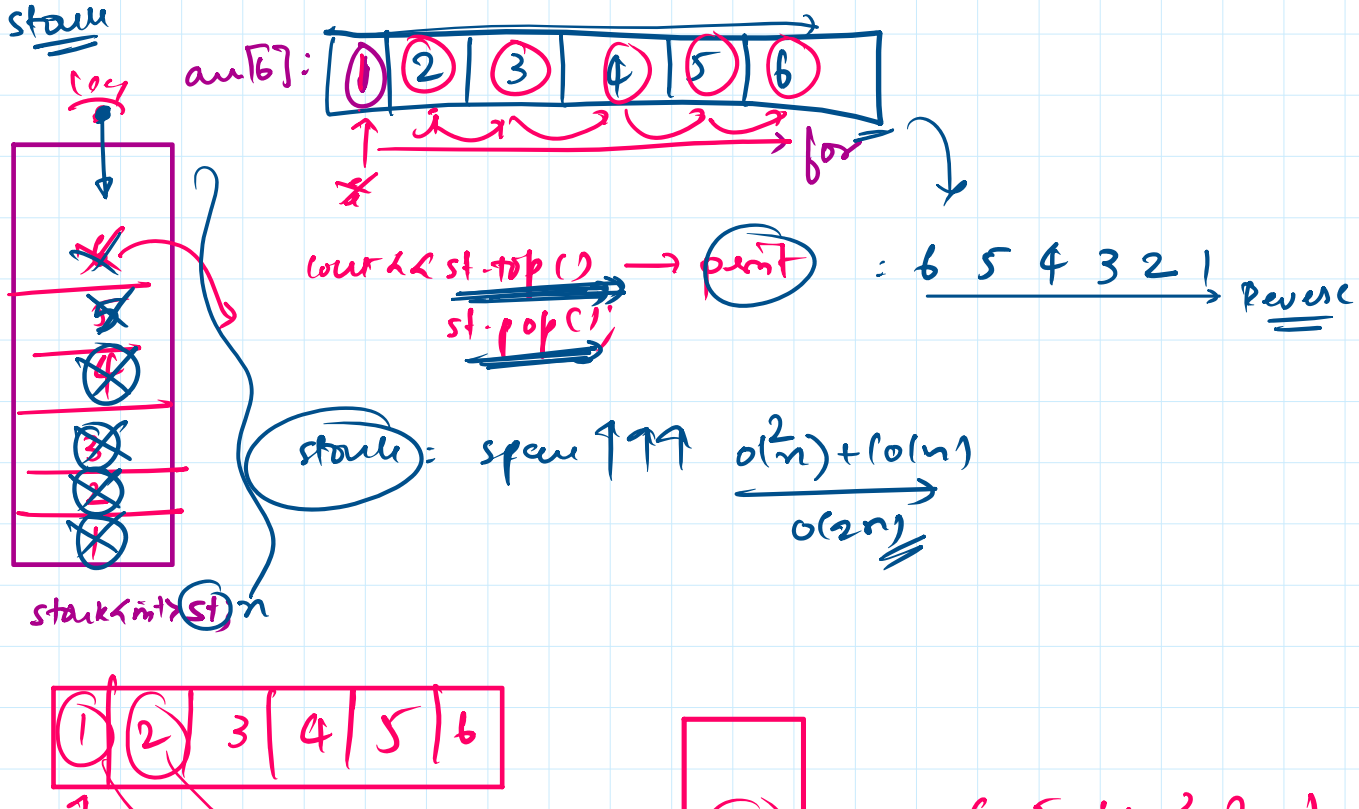
stack

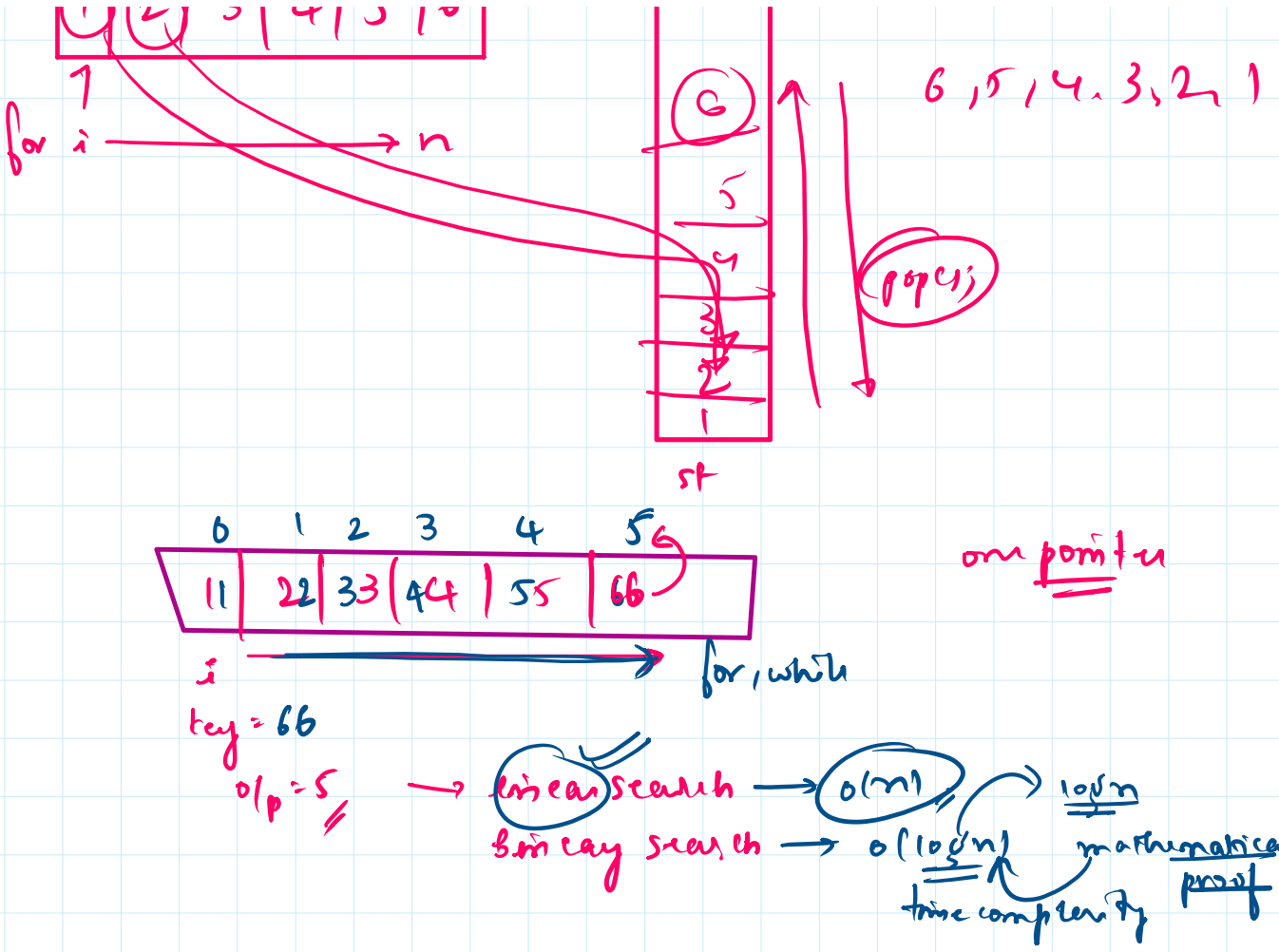






* Reverse an Array using stack;





one pointer

key = 66

o/p = 5

→ linear search → O(n) → log n

binary search → O(log n) → mathematical proof

time complexity

Move Zeroes

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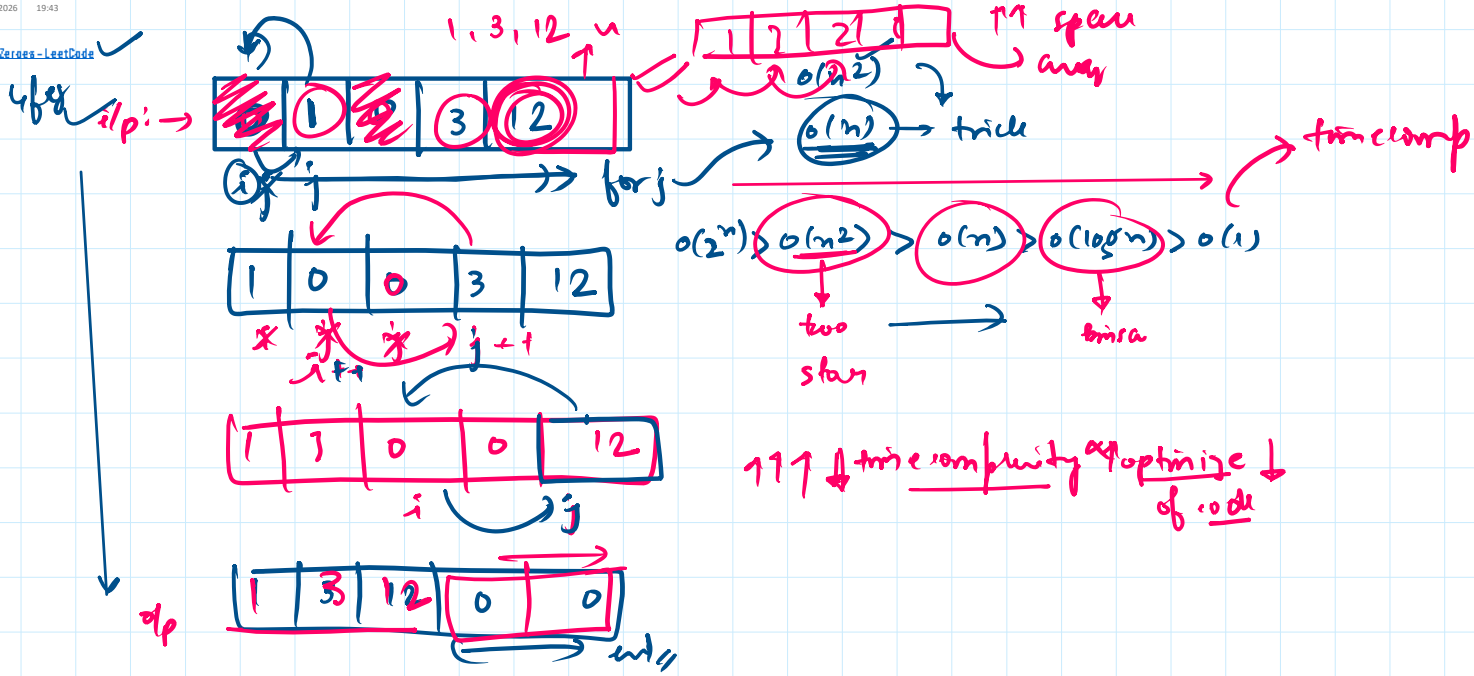
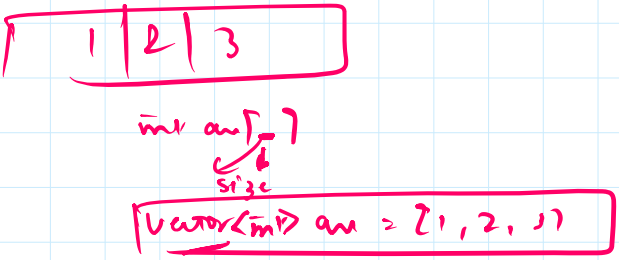
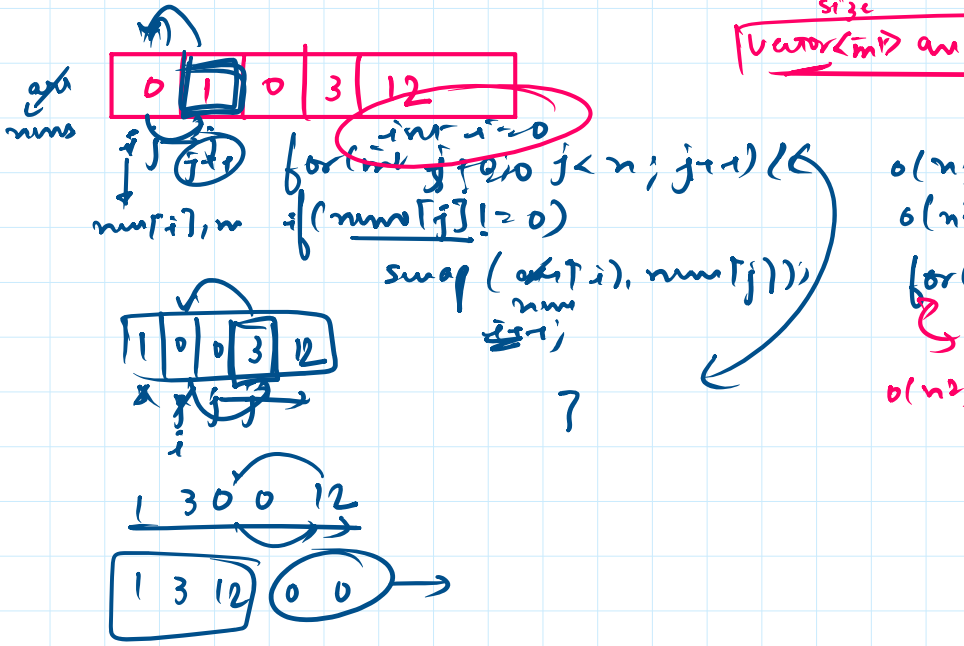
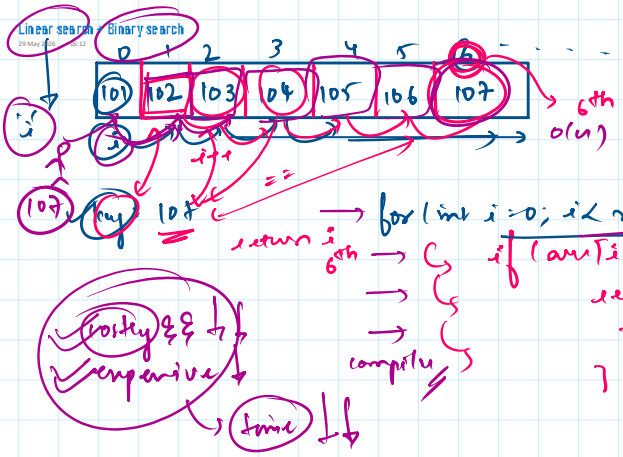
Move Zeroes - LeetCode

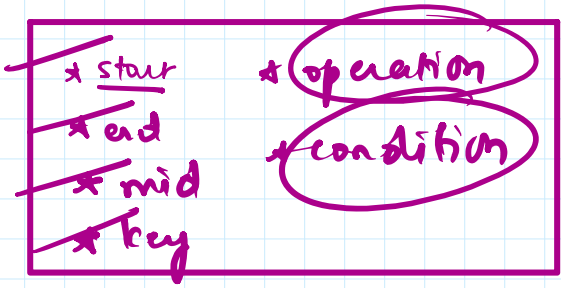


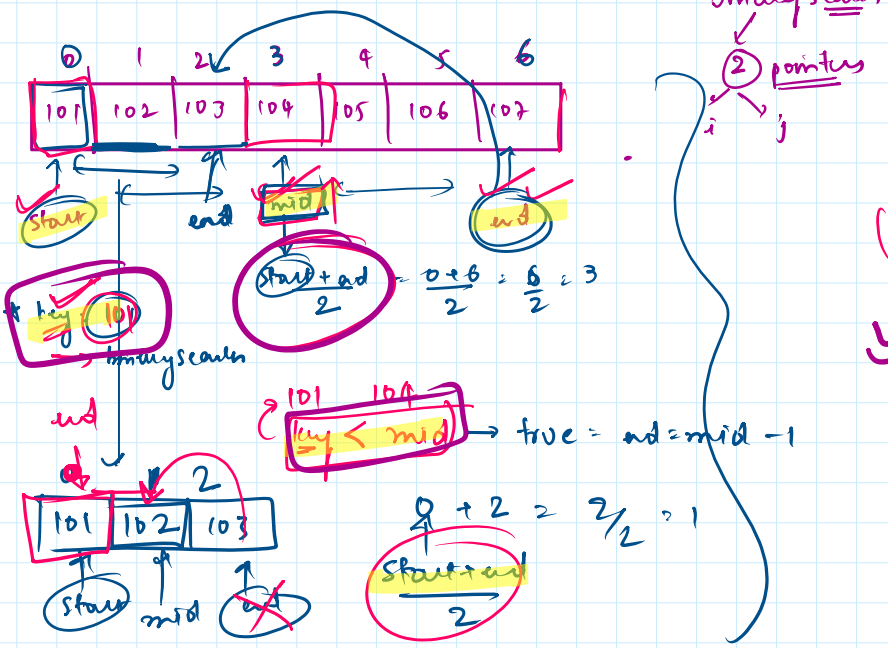


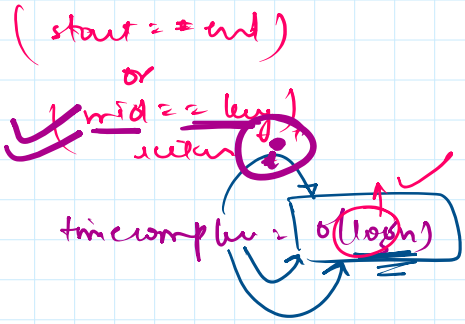


Linear search - Binary search

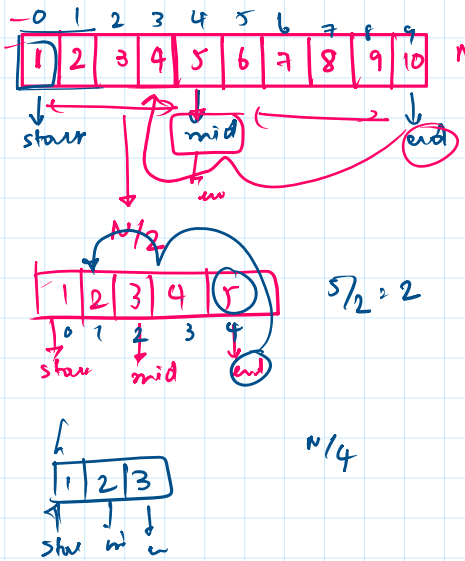


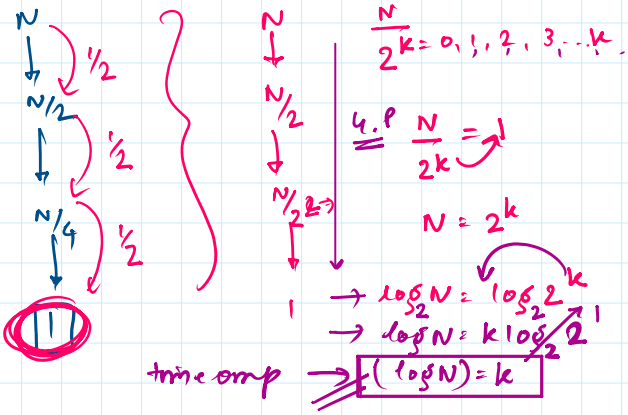




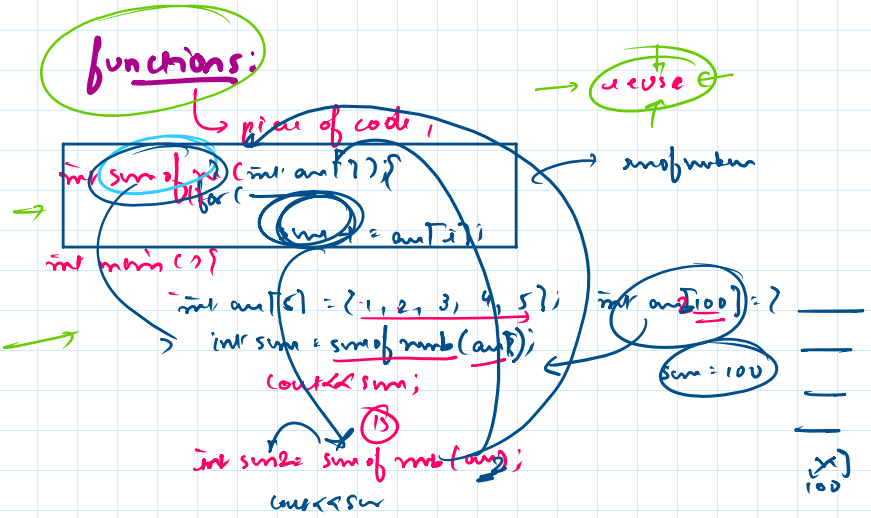


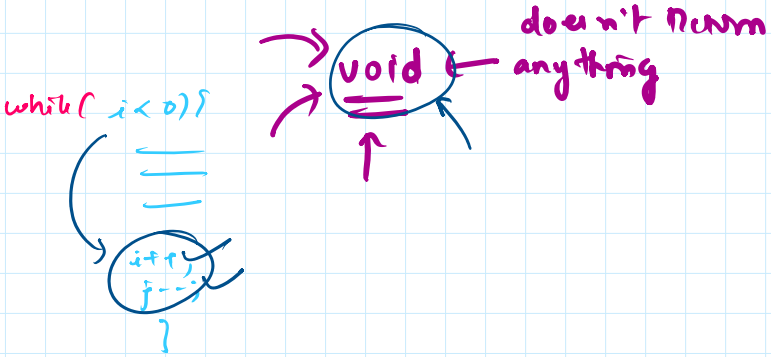
key < mid
101 < 102 → true = end = mid - 1;





$\log_a a = 1$


functions:
piece of code
reuse
sum of numbers
int sum of numb (int arr[7]){
for (
sum = arr[i];
int main (){
int arr[5] = {1, 2, 3, 4, 5};
int sum = sum of numb(arr);
cout<<sum;
15
int sum2 = sum of numb(arr);
cout<<sum
int arr[100] = ?
sum = 100
arr[100]


doesn't return
void
anything
while (i < 0){
i++;
j--;
}

Recursion basis

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→ Recursion.

* function calling itself.

Patterns

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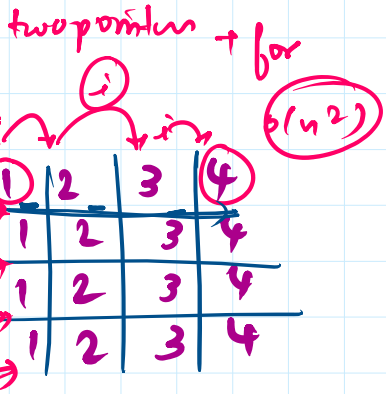
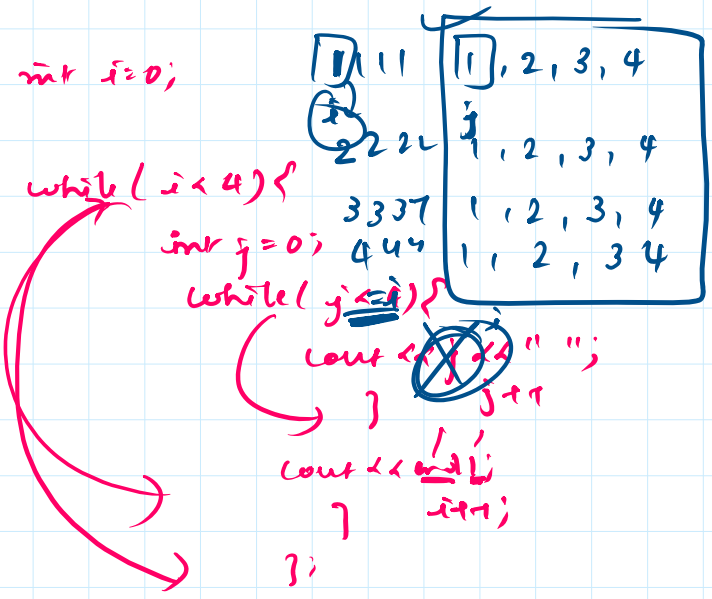
Q.1

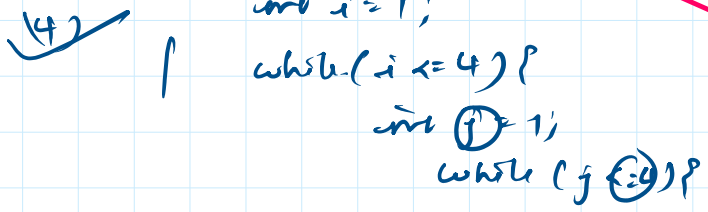
1234
1234
1234
1234

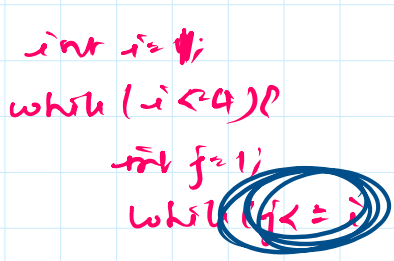


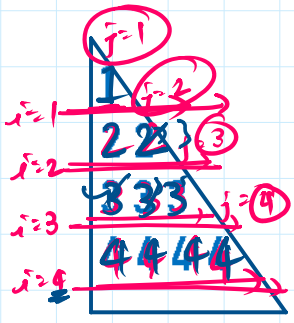


....

1111
2222
3333
4444







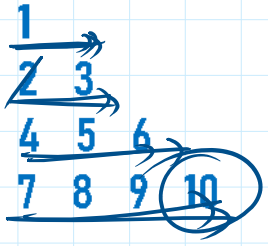
No of row(i) = count
No of col(j)

i	j (j<=i)
1	1 2<=3
2	2 2
3	3 3 3
4	4 4 4 4

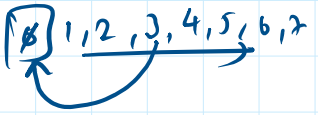
1,2 j<=i → 1 2 3 4
1,2,3
1,2,3,4

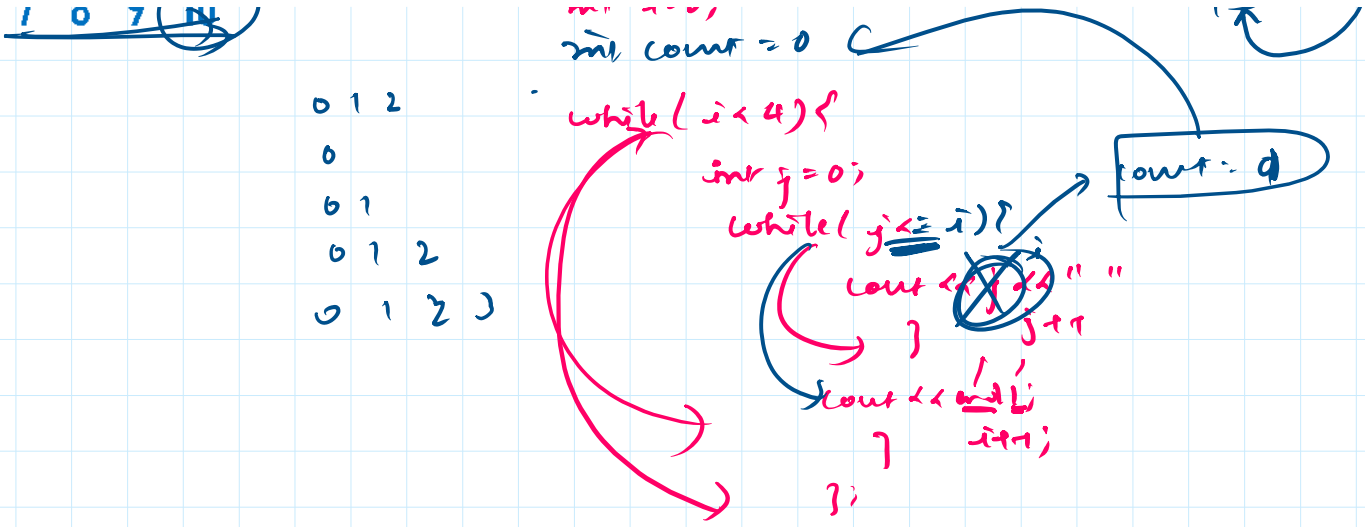
B 1 2 3, 4, 5, 4

count → for j while j



int i=0;
int count=0





0 1 2
0
0 1
0 1 2
0 1 2